**FORMATIVE ASSIGNMENT 2**

## Research-Informed Sequential Models for NLP and Language Technologies

This course focuses on sequential data and methods for modelling dependencies within sequences. In Assignment 1, you applied these concepts to numerical time-series data. In this assignment, you will apply them to language or speech data, which are also sequential in nature. Text may be represented as sequences of characters, words, subwords, or tokens, while speech may be represented as sequences of audio signals or acoustic features. Your work should therefore focus on models and approaches that are appropriate for learning patterns and dependencies in sequential data.

In this group assignment, you will investigate a real-world NLP or language technology problem using one of the approved challenge datasets. The selected challenges address problems involving African languages, communities, or other African contexts.

Your project should address the following broad question: "**How effectively can sequential modelling approaches be used to address a real-world NLP or language technology problem, and what evidence supports the strengths and limitations of the selected approaches?"**

The objective is not simply to obtain the highest evaluation score. You are expected to understand the problem and its sequential nature, investigate the data, review relevant research, make justified model-selection decisions, conduct experiments, and critically analyse your findings.

Each group will work on a different approved challenge. Each group will use one of the following datasets:

- Swahili Audio Classification
- Gender-Based Violence Tweet Classification Challenge 2025
- Sentiment Analysis: To Vaccinate or Not to Vaccinate
- Google NLP Hack Series: Swahili Social Media Sentiment Analysis Challenge
- Swahili News Classification Challenge

### 1. Problem and Data Investigation

Begin by developing a clear understanding of the real-world problem represented by your assigned challenge. Introduce the problem and explain its practical relevance. Discuss the dataset and identify the aspects of the data that are most important to the development of a sequential modelling solution.

Your exploratory analysis should be appropriate to your particular challenge. Depending on the data, relevant issues may include sequence or document length, class distribution, vocabulary characteristics, language variation, noisy data, class imbalance, rare words, spelling variation, code-switching, temporal or sequential dependencies, or other task-specific characteristics.

For speech data, you may investigate characteristics relevant to the representation and modelling of audio sequences.

You are not expected to perform every possible analysis. Instead, select analyses that help you understand the data and provide evidence that may inform your modelling decisions.

Present your findings and explain what they reveal about the problem and how they may influence the design or selection of your models.

## 2. Related Work and Sequential Model Selection

Conduct a focused review of relevant research on your selected problem and closely related applications. Where relevant, consider research involving African languages, low-resource settings, multilingual NLP, speech or language technologies, and **methods for modelling sequential data**.

The purpose of this review is to understand how similar problems have been approached and to use existing research, together with your investigation of the data, to inform your methodological decisions.

Based on your literature review and data analysis, select **five substantially different approaches** that you consider appropriate for modelling the sequential nature of your dataset.

At least **three of the selected approaches must be neural architectures designed to learn dependencies within sequential data**. The remaining two approaches may be alternative approaches that you can justify as a meaningful baseline or comparison.

The three approaches should be sufficiently different to support a meaningful comparison. Minor variations of the same architecture, such as changing the number of layers, hidden units, or other hyperparameters, will not be considered different approaches.

For each selected approach, explain why it was included in your study. Your justification should be informed by the characteristics of your data, relevant findings from previous research, and your understanding of the strengths and limitations of the approach.

For example, if you select a particular architecture because it is expected to capture long-range dependencies, contextual relationships, local patterns, or other characteristics relevant to your sequential data, explain the basis for this decision and support it with relevant literature where appropriate.

Existing research should inform your decisions, but your objective is not to reproduce an existing study. Your own experiments should provide the evidence for the conclusions you draw.

## 3. Experimental Investigation

Using the same five approaches selected and justified through your related work and model-selection process, design, implement, and evaluate your solution.

Clearly describe your experimental methodology. This should include the relevant data preparation and preprocessing, sequence representation or feature extraction where appropriate, the construction of model inputs, training, hyperparameter tuning, validation strategy, and the main design decisions made during implementation.

Evaluate all five approaches using evaluation metrics appropriate to your specific challenge. Clearly state and justify your choice of evaluation metrics. Present your results using appropriate tables and figures(learning curves, ROC curves, confusion matrices, etc). Compare the performance of the three approaches and discuss the differences observed.

Your discussion should go beyond identifying the approach with the highest score. Interpret the results and, where appropriate, relate differences in performance to the characteristics of the data, the ability of the models to capture relevant sequential patterns, differences in their architectures, and findings from previous research.

### 4. Error Analysis and Limitations

A strong evaluation requires more than reporting numerical results. Analyse meaningful cases in which your models perform poorly. The form of this analysis will depend on your challenge. For text-based problems, you may examine incorrectly classified examples, ambiguous language, unusual vocabulary, spelling variation, code-switching, or other difficult linguistic patterns. For speech-based problems, investigate difficult audio samples or characteristics that may contribute to incorrect predictions.

Present evidence from your results and discuss possible reasons for the observed errors. Your analysis should demonstrate an understanding of the limitations of your approaches and of the challenges presented by the data.

Conclude by discussing the main limitations of your investigation and proposing realistic directions for improvement or future work.